Tracking Pomona's Forward Food Pledge: Measuring the Shift to Plant-Based Protein in Dining Purchases

Data Science Capstone, Pomona College

First Outline (Tentative) · October 2026

1 Introduction

Introduce Pomona's Forward Food Pledge (a student-led commitment to make 50% of protein purchases plant-based by 2030). Set up and explain my research question: what share of Pomona's protein purchasing is plant-based, how has it changed, and what would it take to reach 50% by 2030?

2 Background

How the pledge works and why protein is the focus. Institutions can set a menu goal or a procurement goal. Pomona's pledge is a procurement goal for protein. I'll explain why shifting protein matters (animal protein has a much larger environmental footprint than plant protein).

3 Data

I'll describe the Dining purchase records from the Office of Sustainability which includes about 20 vendors for FY24–FY26, plus earlier years if available. I'll explain what each record contains and what it can't tell us. Most records are yearly totals, not individual orders.

4 Defining and Classifying Protein

I'll explain what counts as protein, including how to handle dairy, eggs and mixed items according to the details of the pledge. Then I'll describe how I combine every vendor's file into one table and classify each product as animal protein, plant protein, or non-protein. I'll check a sample by hand to measure accuracy.

5 Measuring Progress

I'll calculate the plant-based share of protein purchases for each year, by both dollars and weight, since the two can tell different stories. I'll compare against the pledge baseline and project the trend to 2030 to show whether Pomona is on track. I'll also break the share down by vendor, by dining hall, and by which proteins are most impactful.

6 Dashboard

This section presents the dashboard for Dining and the Sustainability Office. It shows pledge progress, the protein mix over time, and where the biggest opportunities for change are. I'll also describe the pipeline behind it which will allow subsequent year's data to be processed much faster (add the files, reviews any new products, and the dashboard updates).

7 Food Waste (if raw Metafoodx data is available)

If I get the raw Metafoodx data, I'll check whether plant-based protein dishes are wasted more or less than animal protein dishes. That matters because buying more plant protein only helps if people eat it. If the raw data isn't available, I'll leave this as future work.

8 Ethical Considerations

- What counts as protein and whether to measure by dollars or weight can make Pomona look closer to or further from its goal.
- Halls serve different menus and different numbers of people, so I'll avoid comparisons that unfairly blame staff at one hall.
- Some pricing information maybe shouldn't be shown publicly.
- The pledge could limit what students are able to eat, and I'll discuss that tension.

9 Limitations and Future Work

The main limitations are that most purchase data are yearly totals, weights are missing for many vendors, and some products are hard to classify. I'll also recommend changes Dining could make, such as tagging plant proteins when purchases are recorded and getting dated orders from vendors.

10 Conclusion

I'll summarize where Pomona stands on its Forward Food Pledge, what's driving progress or holding it back, and how the dashboard and pipeline can support tracking through 2030.